CODEALPHA

PYTHON PROGRAMMING INTERNSHIP

PROJECT REPORT

TASK 2: STOCK PORTFOLIO TRACKER

A Simple Stock Investment Calculator Using Python

Submitted By:

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Branch:

Computer Science Engineering

Internship Domain:

Python Programming

Organization:

CodeAlpha

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Project Type:

Internship Task – 2

(Stock portfolio tracker)

DECLARATION

I hereby declare that the project entitled "Stock Portfolio Tracker" has been developed by me as part of the CodeAlpha Python Programming Internship – Task 2.

This project was developed using Python programming concepts and follows the requirements provided for the internship task. The application allows users to enter stock names and quantities, uses a predefined dictionary containing stock prices, and calculates the total investment value.

I declare that the work presented in this report is my own work carried out during the internship period for educational and learning purposes.

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Signature:svk

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to CodeAlpha for providing me with the opportunity to participate in the Python Programming Internship.

I am thankful for the practical tasks provided during the internship, which helped me apply Python programming concepts to develop useful applications.

I would also like to thank my teachers, mentors, friends, and family for their guidance, encouragement, and support throughout the development of this project.

The Stock Portfolio Tracker helped me improve my understanding of Python dictionaries, user input, basic arithmetic operations, conditional statements, and file handling.

Finally, I competed this project, and I gain new knowledge experience etc.

ABSTRACT

The Stock Portfolio Tracker is a simple Python-based application developed as part of the CodeAlpha Python Programming Internship – Task 2.

The main purpose of the project is to calculate the total investment value of stocks based on manually defined stock prices. The application uses a hardcoded Python dictionary containing stock symbols and their corresponding prices.

The user enters the stock name and quantity they own. The program retrieves the corresponding stock price from the dictionary and calculates the investment value using basic arithmetic. The total value of all entered stocks is then displayed to the user.

The project demonstrates fundamental Python concepts including dictionaries, input/output operations, variables, conditional statements, loops, and basic arithmetic. File handling can also be used to save the portfolio results in a text or CSV file.

This project provides practical experience in developing a simple financial calculation application using Python.

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1. INTRODUCTION

A stock portfolio consists of different stocks owned by an individual or organization. Tracking the value of these investments manually can become difficult when multiple stocks are involved.

The Stock Portfolio Tracker is a simple Python application developed to calculate the total investment value based on stock prices and quantities entered by the user.

The project uses a hardcoded dictionary to store stock symbols and their corresponding prices. The user provides the stock name and the quantity owned. The application calculates the investment value of each stock by multiplying its price by the quantity.

The values of all entered stocks are added together to determine the total investment value.

This project was developed as part of the CodeAlpha Python Programming Internship and provides practical experience with dictionaries, input/output, arithmetic operations, and basic program logic.

2. PROBLEM STATEMENT

The objective of this project is to develop a simple Stock Portfolio Tracker using Python.

The application should allow users to enter stock names and quantities and calculate the total investment based on predefined stock prices.

The stock prices should be stored in a hardcoded Python dictionary. The program should retrieve the price of the selected stock, calculate its investment value, and display the total investment value.

The project may optionally save the calculated portfolio information in a text or CSV file.

3. OBJECTIVES

The main objectives of the Stock Portfolio Tracker are:

1. To develop a simple stock investment calculator using Python.
2. To store stock prices using a Python dictionary.
3. To allow users to enter stock names.
4. To allow users to enter the quantity of stocks.
5. To calculate the investment value of each stock.
6. To calculate the total investment value.
7. To demonstrate the use of basic arithmetic operations.
8. To practice Python input and output operations.
9. To use conditional statements for validating stock names.
10. To optionally save portfolio results in a text or CSV file.

4. SCOPE OF THE PROJECT

The scope of this project is to create a simple portfolio tracking application for calculating the value of selected stocks.

The application uses manually defined stock prices stored in a Python dictionary. It does not retrieve live stock prices from the internet or use an external financial API.

The user enters the stock symbol and quantity, and the application calculates the investment value using the predefined price.

The project is intended for educational purposes and demonstrates basic programming concepts rather than providing real-time financial market information.

The application can be extended in the future to include live market data, graphical analysis, portfolio history, and advanced financial features.

5. TECHNOLOGIES AND TOOLS USED

5.1 Python

Python is the main programming language used to develop the Stock Portfolio Tracker.

5.2 Dictionary

A Python dictionary is used to store stock symbols and their predefined prices.

For example:

{"AAPL": 180, "TSLA": 250}

5.3 Visual Studio Code

Visual Studio Code was used to write, edit, and execute the Python program.

5.4 Console

The application uses console input and output to communicate with the user.

5.5 File Handling

File handling can optionally be used to save the portfolio results in a .txt or .csv file.

6. SYSTEM REQUIREMENTS

6.1 Hardware Requirements

- Computer or laptop
- Minimum 4 GB RAM
- Keyboard
- Basic processor
- Sufficient storage space

6.2 Software Requirements

- Windows operating system
- Python 3.x
- Visual Studio Code or another Python IDE
- Python interpreter

6.3 Additional Requirements

The basic application does not require an internet connection, external API, or database because stock prices are manually defined in the program.

7. PROJECT DESCRIPTION

The Stock Portfolio Tracker is a simple Python application that calculates the total investment value of stocks owned by the user.

The application contains a dictionary with predefined stock symbols and their corresponding prices. For example, stock symbols such as AAPL and TSLA can be stored with their manually defined prices.

The user enters a stock symbol and the number of shares owned. The program checks whether the entered stock exists in the dictionary.

If the stock is available, its predefined price is retrieved. The investment value is calculated using the following formula:

Investment Value = Stock Price × Quantity

If multiple stocks are entered, the individual investment values are added together to calculate the total portfolio investment.

The final investment value is displayed to the user. The result can optionally be saved to a text or CSV file.

8. FEATURES OF THE APPLICATION

8.1 Predefined Stock Prices

The program contains a dictionary with manually defined stock prices.

8.2 User Input

The user can enter the stock symbol and quantity of shares.

8.3 Stock Validation

The program checks whether the entered stock exists in the predefined dictionary.

8.4 Investment Calculation

The value of each stock is calculated by multiplying its price by the quantity.

8.5 Total Investment

The application calculates and displays the total value of all entered stocks.

8.6 Multiple Stocks

The user can enter more than one stock and calculate the combined investment value.

8.7 Optional File Saving

The portfolio result can optionally be stored in a text or CSV file.

8.8 Simple Interface

The application uses basic console input and output, making it easy to use and understand.

9. METHODOLOGY

The development of the Stock Portfolio Tracker was completed through the following steps:

Step 1: Define Requirements

The requirements of the CodeAlpha task were analyzed.

Step 2: Create Stock Dictionary

A Python dictionary was created containing stock symbols and their predefined prices.

Step 3: Accept User Input

The program asks the user to enter the stock name and quantity.

Step 4: Validate Stock

The entered stock symbol is checked against the dictionary.

Step 5: Retrieve Stock Price

If the stock exists, its corresponding price is retrieved.

Step 6: Calculate Investment

The investment value is calculated by multiplying the stock price by the quantity.

Step 7: Calculate Total

The values of all selected stocks are added to calculate the total investment.

Step 8: Display Result

The total investment value is displayed to the user.

Step 9: Optional File Saving

The portfolio information can optionally be saved to a text or CSV file.

10. SYSTEM WORKFLOW

The basic workflow of the Stock Portfolio Tracker is:

START

↓

Create stock price dictionary

↓

Ask user for stock name

↓

Ask user for quantity

↓

Check whether stock exists

↓

Stock available?

YES → Retrieve stock price

NO → Display invalid stock message

↓

Calculate:

Price × Quantity

↓

Add investment value to total

↓

Display total investment

↓

Optionally save result

↓

END

11. ALGORITHM

1. Start the program.
2. Create a dictionary containing predefined stock prices.
3. Initialize the total investment value to zero.
4. Ask the user to enter the stock symbol.
5. Ask the user to enter the quantity.
6. Check whether the stock symbol exists in the dictionary.
7. If the stock exists, retrieve its price.
8. Multiply the stock price by the quantity.
9. Add the calculated value to the total investment.
10. Repeat the process if the user wants to enter another stock.
11. Display the total investment value.
12. Optionally save the portfolio details to a text or CSV file.
13. End the program.

12. PYTHON CONCEPTS USED

12.1 Dictionary

A dictionary is used to store stock symbols and their corresponding prices.

Example:

stocks = {"AAPL": 180, "TSLA": 250}

The stock symbol acts as the key and the price acts as the value.

12.2 Input and Output

The input() function is used to receive stock names and quantities from the user. The print() function is used to display the results.

12.3 Basic Arithmetic

Multiplication is used to calculate the investment value.

Investment Value = Price × Quantity

Addition is used to calculate the total investment.

12.4 Variables

Variables are used to store stock names, quantities, prices, individual investment values, and the total investment.

12.5 Conditional Statements

if-else statements can be used to check whether a stock exists in the dictionary.

12.6 Loops

A loop can be used to allow the user to enter information for multiple stocks.

12.7 File Handling

If implemented, file handling can be used to save the portfolio results in a .txt or .csv file.

13. CALCULATION METHOD

The Stock Portfolio Tracker uses a simple mathematical calculation to determine the value of the user's investment.

Formula

Investment Value = Stock Price × Quantity

For example, if the predefined price of a stock is ₹180 and the user enters a quantity of 5:

Investment Value = 180 × 5

Investment Value = ₹900

If the user has multiple stocks, the values of all stocks are added together.

For example:

Stock 1 Investment + Stock 2 Investment = Total Investment

This calculation provides the total value based on the manually defined prices in the program.

14. IMPLEMENTATION

The Stock Portfolio Tracker was implemented using Python.

First, a dictionary is created containing predefined stock symbols and prices. The program then accepts stock names and quantities from the user.

The entered stock symbol is checked against the dictionary. If the stock exists, the program retrieves its price.

The investment value is calculated by multiplying the price by the quantity entered by the user.

If multiple stocks are entered, each investment value is added to a total variable.

Finally, the program displays the total investment value.

The implementation follows the simplified requirements of the CodeAlpha task and focuses on dictionaries, input/output, arithmetic operations, and optional file handling.

15. TESTING AND RESULTS

Testing was performed to verify that the Stock Portfolio Tracker produces the expected results.

Test Case	Input/Action	Expected Result	Result
1	Start program	Program starts successfully	Pass
2	Enter valid stock	Stock price is retrieved	Pass
3	Enter quantity	Quantity is accepted	Pass
4	Enter valid stock and quantity	Investment value is calculated	Pass

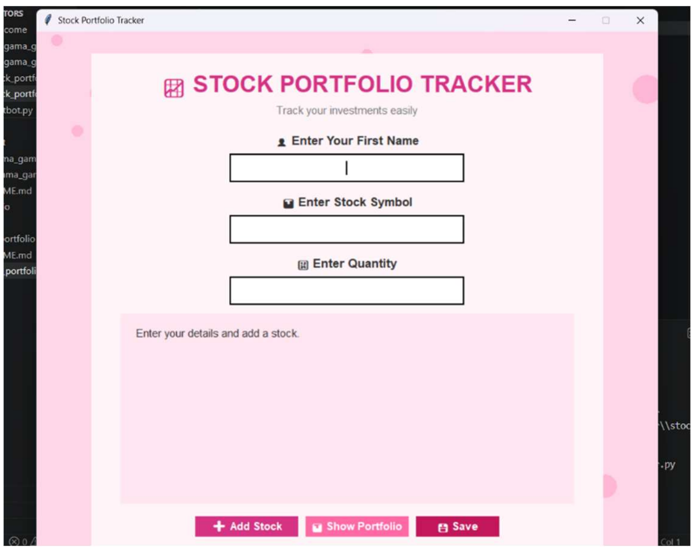
Test Case	Input/Action	Expected Result	Result
5	Enter multiple stocks	Total investment is calculated	Pass
6	Enter invalid stock	Appropriate message is displayed	Pass
7	Save result	Portfolio information is saved	Pass

The testing results indicate that the main functionality of the Stock Portfolio Tracker works as expected.

16. OUTPUT SCREENSHOTS

16.1 Program Starting Screen

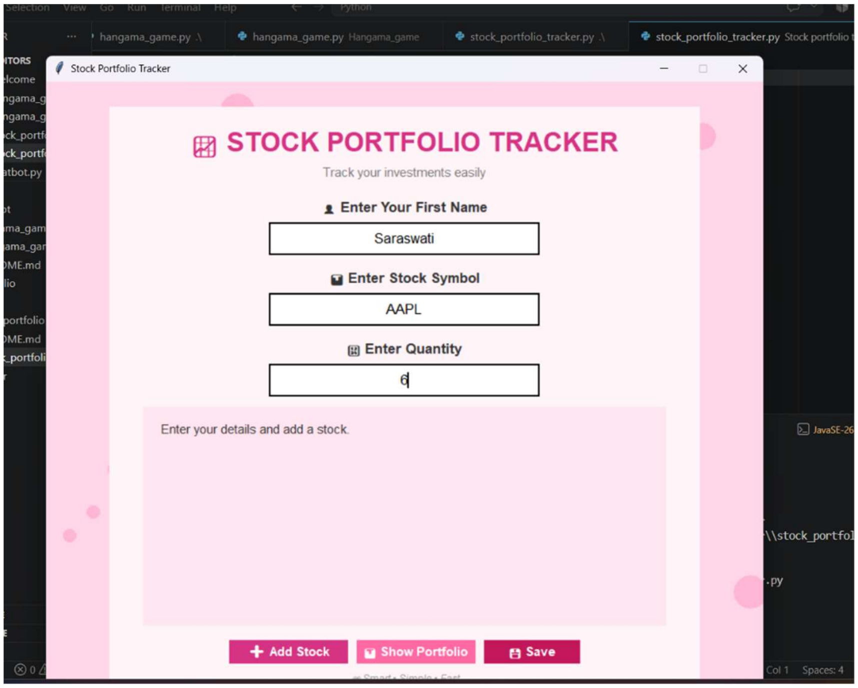
Figure 1: Stock Portfolio Tracker – Starting Screen



The above screenshot shows the program after execution and displays the options or instructions for entering stock information.

16.2 Entering Stock Name

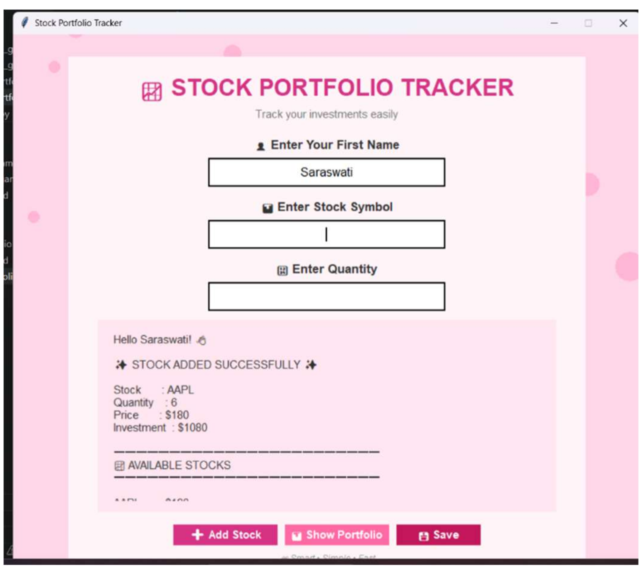
Figure 2: Entering Stock Symbol



The above screenshot shows the user entering the stock symbol.

16.3 Entering Quantity

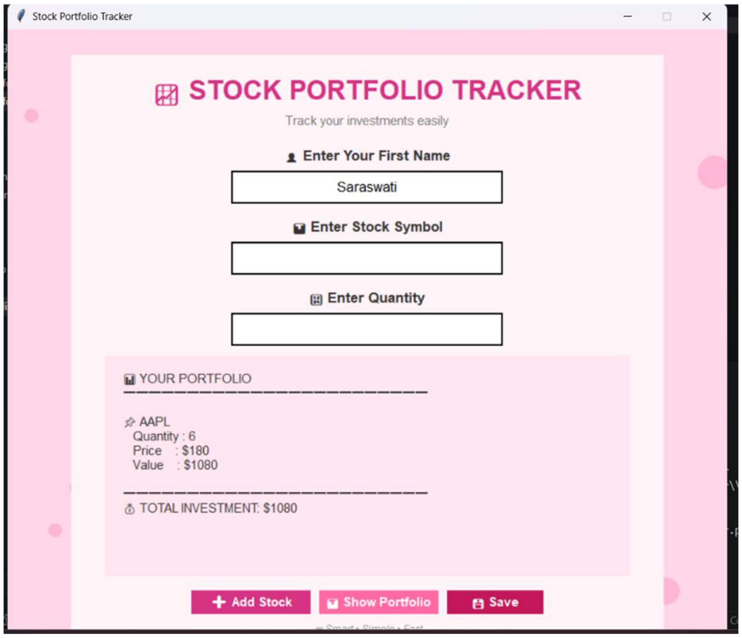
Figure 3: Entering Stock Quantity



The above screenshot shows the quantity of shares entered by the user.

16.4 Investment Calculation

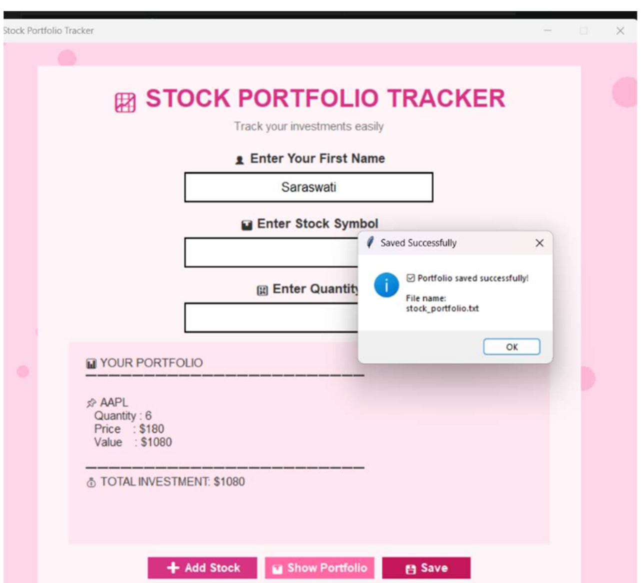
Figure 4: Individual Investment Calculation



The above screenshot shows the calculated investment value based on the stock price and quantity.

16.5 Total Investment

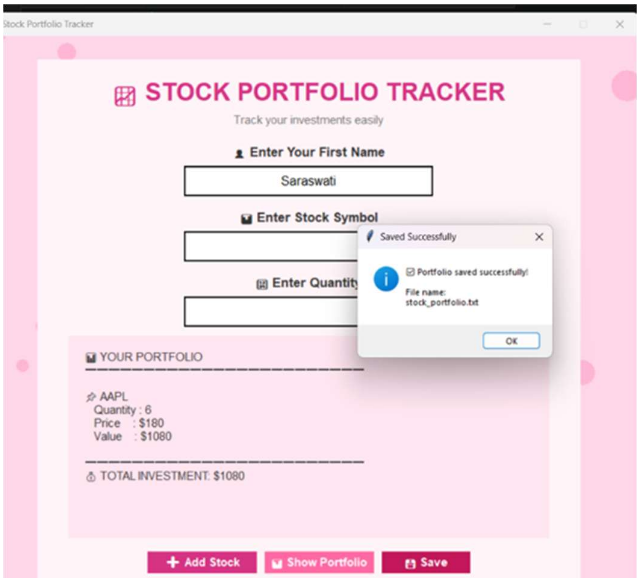
Figure 5: Total Portfolio Investment



The above screenshot shows the total investment value calculated by the program.

16.6 Saved Result

Figure 6: Saved Portfolio Result



17. ADVANTAGES

The main advantages of the Stock Portfolio Tracker are:

- Simple and easy to use.
- Easy to understand for beginners.
- Demonstrates practical use of Python dictionaries.
- Performs investment calculations quickly.
- Supports multiple stocks.
- Does not require an internet connection.
- Does not require a database.
- Does not require a live stock market API.
- Can optionally save results to a file.
- Can be easily extended with additional features.

18. LIMITATIONS

The current version of the Stock Portfolio Tracker has some limitations:

1. Stock prices are manually defined.
2. The application does not provide real-time stock prices.
3. The prices are not automatically updated according to the stock market.
4. The project does not connect to a financial API.
5. The application provides basic portfolio calculations only.
6. Advanced financial analysis is not included.
7. The current version does not provide graphical charts.
8. The portfolio data may not be permanently stored unless file handling is implemented.

19. FUTURE ENHANCEMENTS

The Stock Portfolio Tracker can be enhanced in several ways.

19.1 Real-Time Stock Prices

An external stock market API could be integrated to retrieve current stock prices.

19.2 Database Integration

A database could be used to permanently store portfolio information.

19.3 Graphical User Interface

A graphical interface could be developed to make the application more interactive.

19.4 Portfolio Charts

Charts and graphs could be added to visually represent the distribution of investments.

19.5 Profit and Loss

The application could calculate the difference between purchase price and current market price.

19.6 Multiple Portfolio Support

Users could create and manage multiple portfolios.

19.7 Export Options

Portfolio information could be exported to CSV, Excel, or PDF formats.

19.8 Transaction History

The application could maintain a history of stock purchases and sales.

20. CONCLUSION

The **Stock Portfolio Tracker** was successfully developed using Python as part of the **CodeAlpha Python Programming Internship – Task 2**.

The project successfully implements the main requirements of the task by using a hardcoded dictionary containing stock prices and allowing the user to enter stock names and quantities.

The application calculates individual investment values using basic arithmetic and calculates the total investment value of the portfolio.

Through this project, practical knowledge of Python dictionaries, variables, input/output operations, conditional statements, loops, arithmetic operations, and optional file handling was gained.

The project also improved logical thinking and problem-solving skills by applying Python programming concepts to a practical investment calculation problem.

Although the current application is simple and uses manually defined prices, it provides a strong foundation for developing a more advanced stock portfolio management system in the future.